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Article

Moving with method: using cards in movement-based design

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Abstract

Movement-based design methods have gained increased attention across various research fields and practices, particularly in interaction design. By engaging the body in movement, these methods have the potential to explore a richer, more intuitive, and immersive user experience. A recent project MeCaMinD[1] brought together researchers from interaction and sports design to explore, collect, and document movement-based methods and make them practically useful across domains. The methods were explored through a series of workshops, furthering the team's understanding of their use and practical applicability. This understanding was compiled into a set of design cards that we present here. We discuss the experience of using the resulting cards in movement-based design sessions. We found that while the cards are mostly used in designing, planning, and preparing sessions, they also function as support during the design sessions, used by both facilitators and participants. Based on a final ideation session with both novice and experienced facilitators, we sketch ways to support managing the cards during sessions, integrating them with movement and physical action.

RESEARCH HIGHLIGHTS:

- Movement-based design methods were collected and made available in the form of movement-based method cards.
- The design rationale and design process behind the card design is described.
- The use of the cards is discussed, highlighting how they primarily are used for planning design sessions.
- A speculative design session is described, which was focused on how the cards can be meaningfully included in design sessions.

Keywords: *movement-based design methods; design cards; movement-based interaction; sports and health; movement-based design*

1 Introduction

Movement-based design methods (Vega-Cebrián et al., 2023) combine creative design practices with participants engaging with the whole body. Such methods have gained increased attention across various research fields and practices, particularly in interaction design. Methods find applications ranging from developing body-worn technology and refining training techniques in health and sports to crafting artistic performances and playground designs. Their broad relevance lies in their capacity to deepen understanding of contextual, social, practical, and experiential dimensions of design while also sparking creativity. Many of these methods can be utilized early in projects, providing valuable insights into the embodied experiences of stakeholders.

A recent international project brought together researchers from interaction design and sports design. The project focused on collecting, documenting, and exploring movement-based design methods to enhance their accessibility and practical utility across diverse domains. Its main outcome was a set of design cards. These summarize tested and tried methods and are intended to guide and reflect knowledge relevant to the design of movement-based experiences. Generally, design cards are a common way to support design processes (Halskov and Dalsgård, 2006; Lucero et al., 2016) for aiding ideation and evaluation and can serve as a dissemination of academic knowledge into practical use. Their physical format allows for organization on a table, facilitating the representation of different design perspectives or stages in the design flow (Lucero et al., 2016).

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However, it is not obvious that cards are a suitable format also for movement-based design methods. While many of the advantages that cards can offer apply also in this context, some might be lost due to the essential particularity of these methods, being that designers themselves are in movement. For example, it is unclear whether cards can support design overviews or can be added or removed at different stages of the design process. Moreover, using cards could impact movement and physical action in movement-based design sessions.

In this article, we share the results and practical insights from developing and using method cards to enrich movement-based experiences in more than 20 design sessions. Besides presenting the resulting card set, we focus in particular on a co-design session in which we explored the opportunities and drawbacks of bringing cards into movement-based design sessions. We highlight why and how cards are usefully brought into sessions, and elicit strategies employed to make the most of the cards in the in-session situation. We also discuss various creative solutions for card management, aimed at addressing issues arising from using cards when moving.

2 Background

Movement practices are eminently physical and highly contextual—they are situated (Buchenau and Suri, 2000) and framed by practicalities (Oulasvirta et al., 2003); they are strongly shaped by the participants' former experiences (Loke and Schiphorst, 2018). Designing for such practices requires physical and practical engagement and first-hand experience (Höök et al., 2016a, 2016b; Hummels et al., 2007). Consequently, movement-based design methods have emerged in disparate research domains, with diverse theoretical justifications and methodological approaches. The work reported in this paper is grounded in two such fields: the field of interaction design and that of design for sports technology. In both, movement-based methods have been argued for from a phenomenological stance, stating that our perception of the world is fundamentally grounded in how we inhabit it (Merleau-Ponty, 2013) and that perception, action, and cognition all are necessarily embodied phenomena (Chemero, 2011).

2.1 Movement-based interaction design

Within Interaction Design, several scholars have used movement-based design methods as a resource for the process of designing technology. They emerge in design endeavors that favor a holistic approach to design, where the dynamic interrelation between technology and use evolves over time (Verbeek, 2005). Early work includes the approach of using 'scenarios' (Carrol, 1999; Bødker, 2000; Howard et al., 2002a, 2002b), brief scenes describing user and/or usage situations, to charter the domain and usage for new products. However, early approaches did not always emphasize the first-hand experience of designers themselves moving or acting out scenes, for example, some included actors doing the moving and acting.

More recently, methods of bodystorming (Oulasvirta et al., 2003; Schleicher et al., 2010; Márquez Segura et al., 2016; Wilde et al., 2017) have been proposed as beneficial in early stages of the design process, for coming up with ideas grounded in the embodied experience of designers and other participants. These methods emphasize contextual exploration and ideation more than earlier methods had. Focus has also shifted towards enabling designers to build awareness of the fleeting and immediate movement experience by moving themselves (Höök et al., 2016a, 2016b; Márquez Segura et al., 2016; Schleicher et al., 2010). Such design methods emphasize intimate aspects of the body-technology relation, which contribute to their emphasis on enactment, highlighting the first-person experience of those involved and even support them to experiment with movement changes, like habitual movement patterns (Loke and Robertson, 2013; Wilde et al., 2017). The design program of Soma Design (Höök et al., 2016a, 2016b) is an elaborate example of this. Soma Design foregrounds the felt, first-person perspective on the body as central during design and encourages designers to sensitize themselves to working with their design intentionality turned inwards towards the personal experience.

Movement and enactment are not only important for such an inwards focus, but also support foregrounding physical and social contexts where action takes place. In addition to the felt experience and the interrelation between body and technology, early movement enactment is also important to capture how social activities are conducted and experienced. Issues related to socio-spatial aspects of movement have been considered in technology design, but also in the design of spaces (De Matteis, 2020) (e.g. interior design, urban design, VR design) and activities such as structured training exercises that are performed in groups (Turmo Vidal et al., 2023). In movement-based design, the social and spatial organization of bodies and movement becomes a design material (Márquez Segura et al., 2013; Márquez Segura, 2016) that through movement-based design methods becomes available for designers to explore (Schön, 2008). Thus, movement-based design expands the design space and through this offers creative opportunities to designers. In doing so, the approach can offer insights into diverse and inclusive perspectives of a wide variety of people and challenge normative understandings of embodied experiences.

The social aspect of movement also strongly influences the movement-based design methods themselves, as these tend to be activities enacted by groups of people in one and the same space. Movement-based design methods often capitalize on what has been called the second person perspective on movement (Svanæs and Barkhuus, 2020)—the interpersonal, empathic, body relation where participants orient themselves towards and empathize with each other's movements. From a phenomenological point of view, this can be argued for through Radical Embodied Cognition Science (Chemero, 2011): emotions play a role in 'coloring' our experiences and influencing how we perceive and act.

Movement-based design methods will often direct the participant's focus towards other participants and the group as a whole. For example, design methods can suggest mimicking each other or sharing objects or movement among the participants.

Since movement-based design methods tend to create fleeting experiences and designs, finding ways to document the activity becomes central (Turmo Vidal and Márquez Segura, 2018). Movement-based design methods will often combine methods of documentation that capture multiple points of view of the experience (Svanæs and Barkhuus, 2020). In order to capture the felt experience, participants are encouraged to explore their own movement experience as undisturbed as possible, sometimes by reflecting on it in hindsight. This first-person perspective is complemented by also observing and capturing their movement, e.g. through video, which allows for an analysis of the participants' movement, posture, and orientation in space. Video recordings help capture a third person perspective, as it allows us to observe bodies as objects on par with other objects (Márquez Segura et al., 2016). The third-person perspective helps create an understanding of the interrelation and interaction between technology, artefacts, and movement (Turmo Vidal and Márquez Segura, 2018).

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2.2 Movement-based design of sports and health technology

Movement-based design methods are often employed in the design of training and educational activities (Abrahamson, 2014), often in conjunction with designing training artefacts including interactive technology to support the same, and have shown significant potential across various domains.

In some movement domains, the design of new movement practices is a core element of the domain. Already in the 1920s, Oskar Schlemmer explored how new movements of dancers could be designed through creative use of fantastic costumes that would constrain and inspire them to move away from their formal training in ballet (Centellas and Gallego, 2016; Karpashevich et al., 2018). More recent, New Circus is a creative practice where novel forms of performativity are developed making creative use of equipment (Lavers et al., 2019). Contact Improvisation Dance is a dance practice where the 'design' of new movements happens during the performance, in improvisation triggered by the environment and by other dancers (Torrents et al., 2016). And role-playing, which includes experiencing various identities, is an established technique in, for example, feminist self-defense (Hollander, 2015) and can be used to improve the understanding and experience of functional variation and bodily limitations.

In other domains, movement-based design involves a big role for technology. For example, in physical rehabilitation, movement-based design has been applied to allow patients to perform rehabilitation exercises at home. An example is a system developed to provide real-time feedback and adjustments based on predefined parameters, which has the potential of enhancing the effectiveness of rehabilitation without requiring constant presence of medical staff (Garcia and Navarro, 2014). Another example technology designed for movement-based interaction is the TwinkleBands, designed to support movement imitation and coordination in fast-paced fitness training. This wearable technology provides real-time feedback and enhances the coordination of participants in activities such as aerobics and Bodycombat, demonstrating its effectiveness in improving fitness-training outcomes (Zhu and Turmo Vidal, 2021).

Design methods for movement-based technology often take a holistic perspective, considering how design will affect all of the first-person movement experience, the use of technology, the social situation, and the participant roles (Postma et al., 2022). For example, the concept of intercorporeal biofeedback takes into account a plethora of interactional resources available to participants including but not restricted to technology, facilitating a holistic approach to design (Turmo Vidal et al., 2023). In the domain of elderly care, key findings in balance training suggest that simple, tangible, interactive solutions can effectively improve balance without the need for complex screen-based systems (Subramanian, 2021). In a study focusing on the design of accessible movement-based games, a participatory design process involving visually impaired students showed that using building blocks in game design workshops empowered visually impaired students to create engaging and accessible games (Regal et al., 2020).

Movement-based design methods thus provide useful a useful tool in designing movement, as well as in integrating interactive technology into everyday movements, prioritizing natural interaction and designing technology in its context of use. The approach can enhance the biomechanics of athletes by focusing on natural, embodied interactions (van Rheden et al., 2024) and can support the development of training programs that improve efficiency, power, and overall performance by leveraging insights into biomechanics as well as interactive system design (Mueller and Isbister, 2014).

2.3 Categorizing movement-based design methods

Andersen et al. (2020) analyzed 23 methods from academic literature and suggested a typology for movement-based design methods. Their suggested categorization relies primarily on the methods' purpose and role in the design cycle. While this might be considered the most crucial difference between methods, many design methods can be applied for multiple purposes and in different project stages. Furthermore, many methods can be more widely applied than in their original conception with minor modifications.

In order to gain further insights into the variability of methods, Vega-Cebrián et al. (2023) collected expert accounts of the practical use of movement-based design methods in 12 different projects. They documented 41 accounts of design method use, including the purpose, approach, and outcome of using the methods. Based on these accounts, the authors developed a multi-dimensional characterization of methods as 17 categories grouped into five groups: design resources, activities, delivery, framing, and context. The paper explores the first group, design resources, in detail, further characterizing and illustrating how movement, space, and objects can be used as design resources and providing actionable pointers for novice and expert designers.

3 Methodology

The process of developing the cards was a research-through-design (Zimmerman et al., 2007) process. The core team consisted of expert academics in the field who had both theoretical insights into the underlying principles of movement-based design methods and practical experience in engaging with, and teaching, movement-based design. The process fluctuated between phases where the experts themselves were the primary informants and designers using methods and cards, and empirical phases when initially the methods, and later the cards themselves, were put to use in collaboration with other practitioners from the domains of sports technology, sport science, physiotherapy, and interaction design. Some of the practical sessions were experiments, others were teaching events where movement-based design was introduced to students and practitioners, and some were run in authentic research projects.

The process for designing and organizing the organization went through the following stages.

1) Methods were collected, trialed, and organized according to their scope and purpose, as well as their practical requirements. This is discussed in more detail below.
2) The layout and content of the cards were designed to allow for easy reference in planning and executing sessions while providing enough detail on the methods to allow participants to run them. Furthermore, the cards needed to provide links to more in-depth information about the methods and their practical execution.
3) The cards were used, studied, and evaluated in more than 15 separate design sessions (see Appendix 1), including both teaching events (teaching movement-based design to students and practitioners) and design sessions in authentic design projects. One such session, a summer school run towards the end of the project, is detailed in this article.
4) After extensive testing in multiple domains, a reflection process was carried out among the participating experts to

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surface common characteristics of card use. Based on this reflection session, an ideation session was organized with the purpose of surfacing ways in which facilitators can be supported in bringing cards into sessions.

The final card set can be downloaded from https://mecamind.eu/.

3.1 The 4 M design framework

A key consideration for designing the cards was how to bridge between the phenomenological understanding of the body as a feeling, moving, and perceptualizing condition of experience, and the cognitive, abstract, and, to a large extent, textual representation that the movement-based method cards would provide. The 4 M framework (Elbæk et al., 2022), structured by theories of embodied cognition and practical considerations in movement-based design, was used to create the overarching typology for the cards. This framework suggests considering the following elements in structuring a design session:

- The ***movement concepts*** that are targeted in the design exercise and that guide its focus and goal.
- The ***mood*** that participants need to be in to focus on the specific aspects of the movement-based design activity.
- The movement-based design ***methods*** in use, and finally
- The way that the design activities can be ***modified*** on the fly while they are being carried out.

3.2 Selecting and modifying methods

The methods considered for inclusion in the card set categories 'Mood Setters' and 'Movement-Based Design Methods' were primarily drawn from the method collections presented in Andersen et al. (2020) and Vega-Cebrián et al. (2023) (see Background), complemented with some that the participating experts provided but were not covered in previous work. To test the methods in practice, design sessions were run in which selected methods were used. A primary consideration for these sessions was to investigate the practical applicability of the methods outside their original domain, as their practical applicability would need to be documented in the cards and too narrowly scoped methods would need to be omitted. For example, methods developed for the domain of physical training were tried out in the domains of dance and physiotherapy[2]. In some cases, the experiences motivated slight changes to the methods compared to how they had been described in the literature. For example, the card for 'Embodied sketching' is a somewhat simplified version of the full complexities of the method as introduced by Márquez Segura et al. (2016). This may make it easier to start using it—but as facilitators and designers get more fluent in using that card, they can refer back to the original paper to understand and use the method in all its nuances.

Subsequently, a workshop was held with project participants with the purpose of organizing the methods into clear categories and identifying key properties that influence their use, based on the above selection process and the classification suggestions in Vega-Cebrián et al. (2023)). A first version of the cards (both format and organization) was designed to reflect the categories and properties emerging from this workshop.

The first version of the card set was trialed in three design sessions, two run by experts in the project and the last one by external facilitators. Both the card layout and content needed to be slightly modified as a result of these sessions. The number of cards also needed to be reduced for printing. Two main considerations determined the final selection of methods to be included: that (1) the methods were reported as useful in practice by the people who ran or participated in the sessions, and (2) the facilitators of those sessions were able to communicate and instruct the methods based on their descriptions. A final effort concerned proofreading the cards for readability and textual references between the cards; this work was divided between the project participants.

[2] Details can be found in previous publications, here withheld preserve anonymity.

3.3 The CardSet

The overarching typology for cards developed was structured based on the 4 M framework introduced above. The card set contains four sets of cards corresponding to the four types of knowledge proposed by the 4 M framework, with the *Movement Design Method* cards being the key resource. A fifth set of cards, the 'Design Flow' cards, were developed to help first-time facilitators get started with selecting suitable design methods and modifiers. The full card collection is printed on waterproof material so that the cards can be used in various sports and movement settings.

3.3.1 *Movement concepts*

The movement concept cards are included in the card set to help practitioners ground their design process in concepts from movement-related theories. Their inclusion is motivated by how design cards have been seen as a way to make theoretical knowledge developed in academia applicable to practical use (Deng et al., 2014; Peters et al., 2021). They articulate specific aspects of movement and movement-based activities that can be important to look out for during a design session, or suggest desirable outcomes of design activities. The concepts selected for the card set were brought in from theories of movement, sport, embodiment, and movement technology, and hence aid in grounding design work in abstract theory of movement as well as practical design considerations (Koskinen et al., 2013). Their main use is during the planning of design sessions.

A total of 38 cards are included in this set. An example **Concept** card is shown in Figure 1. The Concept cards are primarily intended to be used when planning a session. They tend to be text-heavy, as theoretical concepts need textual descriptions. This particular card was selected for inclusion due to how accessibility issues are sometimes overlooked in movement-based design.

One side of the card (Figure 1 left) is devoted to a description of the key concept, and this side also has a QR code linked to more extensive information. It links to a downloadable Excel file holding the text of the cards and the illustration, as well as 1–3 literature references that have been used as essential knowledge for forming the cards. The other side (Figure 1 right) is devoted to key considerations that can be brought into the design session, and suggests methods and modifiers that could be used to explore them. At the bottom of this side, a square map illustrates the kind of guidance the concept provides to a design process (Waern and Turmo Vidal, 2022). Some concepts are primarily *inspiring*, presenting options and ideas. *Sensitizing* concepts helps bridge between different experts in a design team by providing words to phenomena that designers might observe or experience but have no common language for. Some concepts present clear categorizations that can be used to classify or measure specific qualities of an idea or evaluate a proposed solution; these are classified as *analytical* concepts. Finally, the *guiding* concepts, like the one in the Figure 1 card, help direct design towards a particular desirable goal.

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Figure 1. An example Movement Concept card discussing facets of inclusive sports and disability in sport.

3.3.2 *Movement-based design methods*

The movement-based design methods form the core part of the 4 M framework. The box contains 37 cards of this type. These are practical, hands-on activities with some desirable outcome (or several). Most of them are most relevant for a particular phase in a design process, but some can be used in several phases. A major distinction lies between sensitizing activities that primarily sensitize participants to a particular phenomenon, experience, or theory, versus methods that focus on creative design or redesign. While most of the methods originated in academic literature, 10% of the movement methods were drawn from movement-based design practices such as new circus. Activities that primarily sensitize participants can also sometimes be useful as mood setters.

An example **method card** is shown in Figure 2. This specific method is a good illustration of how some movement-based design methods have emerged in multiple fields with large similarities. The card documents a method that was given the name 'What Can I Do With This'. It describes a design exercise in which different objects are brought into the creative design process, encouraging participants to play around with them, disrupt or collaborate with each other, and reflect on the experience. The precise description of the method was adopted from methods used in teaching creative acrobatic physical education activities, which in turn rely heavily on practices adopted from the New Circus movement and acrobatic elements of performance gymnastics. The creative culture of New Circus centers on using equipment (Lavers et al., 2019) that supports creative forms of performativity. Very similar methods for using artefacts in embodied design ideation have also been described in interaction design literature (Márquez Segura et al., 2016). The method card documents the method with sufficient variability so that it can be used in both physical training and interaction design.

As seen in the figure, both sides of the card are used to convey the method. One side (Figure 2 left) is dominated by a name and a unique illustration. On the same side of the card, simple figures illustrate the practicalities of running the method, including the expected duration, the proposed number of participants, and whether the method needs preparation time or materials. At the bottom, the methods are categorized by four common aspects of movement-based design methods: their ability to foster body awareness, exertion, role-playing, and playfulness. The opposite side (Figure 2 right) has more text. Here, focus is placed on the method's intended outcome, possible props to use, step-by-step instructions, suggested variations, and why/when this method would be chosen over other options. At the bottom of the text side, there is a QR code that links to additional information about the method, in the form of a downloadable Excel file holding the text of the cards and the illustration and 1–3 literature references.

3.3.3 *Mood setter methods*

Movement-based design is typically done as group activities, which is why the social dynamics of the group cannot be overlooked. To this purpose, design sessions are often initiated by performing a set of group activities that do not contribute to the design as such but instead help the group come together, establishing a joint mood within the group, contributing to embodied creativity, and sometimes also sensitizing the participants to the desirable design outcomes. The methods included in this part of the card set are not creative per se but help create a joint focus or mood in the group, and warm up the participants physically, mentally, and socially (Márquez Segura et al., 2021). These exercises are typically used before or between design sessions.

The **Mood Setter** card set includes a total of 37 cards documenting this type of methods. They have primarily been taken

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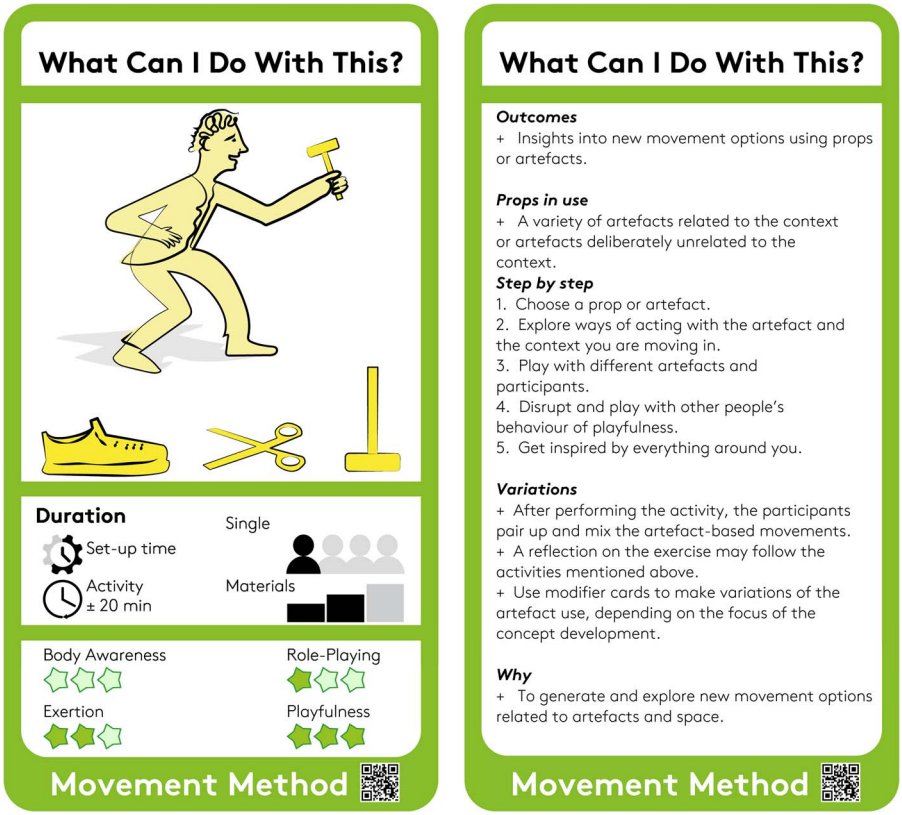


Figure 2. An example Movement Method card, presenting the method 'what can I do with this' derived from creative acrobatics practices.

from the sports, health, and movement domains, or else are common icebreaker and teambuilding activities that were modified for movement, with a focus on offering embodied creativity training potential. The Opposite Moves method illustrated in Figure 3 represents an exercise informed by improvisational dance. Participants collaborate in pairs and are encouraged to ideate on movements that can be interpreted as opposite to what the partner shows. For example, the first mover makes very staccato movements, and the other partner performs a soft and flowful move. Change initiative and the second mover shows, e.g., a movement at ground level, and the first participant performs a movement at high levels, etc. The exercise will lightly warm up the participant's body and train improvisational skills using the body as a creative resource. This particular activity has the potential to be used as an embodied icebreaker. Other activities, such as for example the Shrinking Ship mood setter, instead emphasize teambuilding. In the Shrinking Ship activity, the group uses a rope or blanket to represent the 'ship'. Participants are on a ship that is slowly sinking. Their goal is to stay within the boundaries of the ship as it sinks and thus shrinks by gradually reducing the size of the ship (represented by the rope or blanket). As the space becomes smaller, participants must work together to ensure everyone stays within the boundaries of the ship.

As seen in Figure 3, the Mood Setter cards are organized much like the method cards. One side (Figure 3 left) contains the Mood setter's name and a unique illustration. On the same side of the card, figures illustrate the practicalities of running the method, such as the expected duration, the proposed number of participants, and whether the method needs preparation time or materials and a QR code. A three-star rating at the bottom indicates the extent to which the exercise can help warm up participants physically through *exertion*; help set a *playful mood*; warm up participants in acting *creatively*; and/or create *body awareness*. For most of the mood setter methods, the QR-link links to an instruction video illustrating the execution of the activity. On the opposite side (Figure 3 right), the expected outcome and props required for the activity are described, and a detailed description of instructing the activity is provided. At the bottom, activity variations are proposed, and finally, a reason for using the specific mood-setting activity is given.

3.4 Modifiers

In a review by Peters et al. (2021), it was noted how tools for collaborative ideation will often include some way to disrupt a design process and that a wide variety of tools, props, and instructions can be used for this purpose. The **Modifier cards** (Figure 4) represent such disruptions and are intended to be brought into sessions as challenges and modifications to the design process, fueling divergent thinking to scaffold the process (Peters et al., 2021). They have a plethora of uses: facilitators can use them to instruct participants or hand them out to participants to allow them to self-organize.

The modifier categories were inspired by the people, activity, context, and technology framework from Adikari et al. (2023), complemented with categories based in ports and movement practices and established design methods. The collection of 306 modifier cards is divided into five different categories: modifying movement, modifying structure, modifying people, modifying artefact, and modifying motive). It also contains one set of empty cards for facilitators to add their own modifiers.

The five categories of cards provide a structured foundation for enhancing movement-based design through diverse research insights. The Modifying Movement cards are rooted in fundamental movement skills, including locomotor, non-locomotor, and

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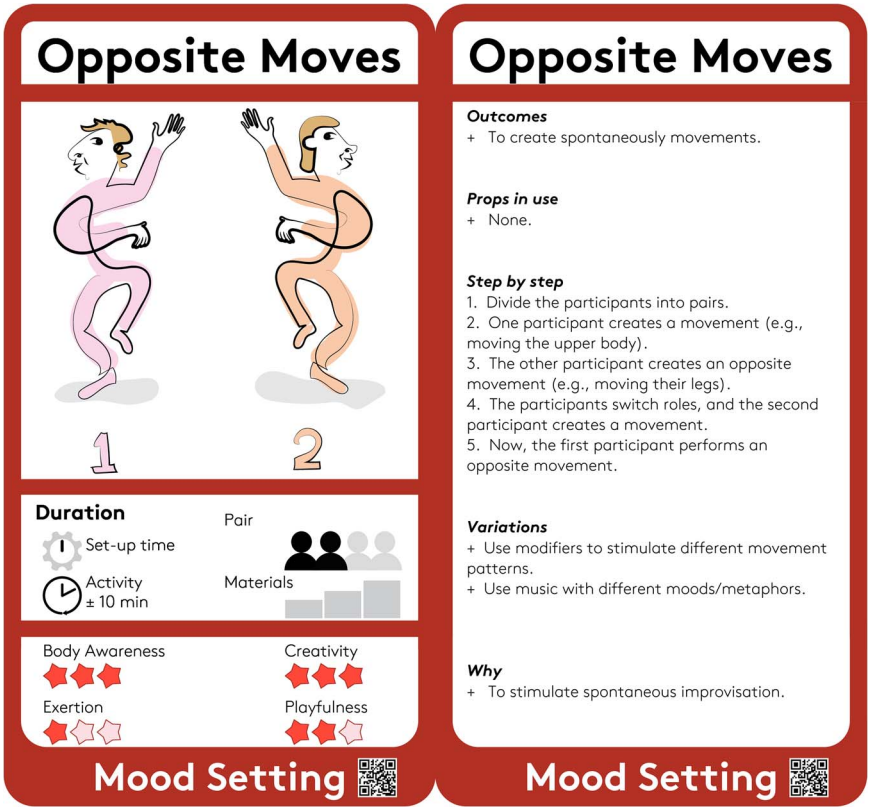


Figure 3. An example Mood Setter card.

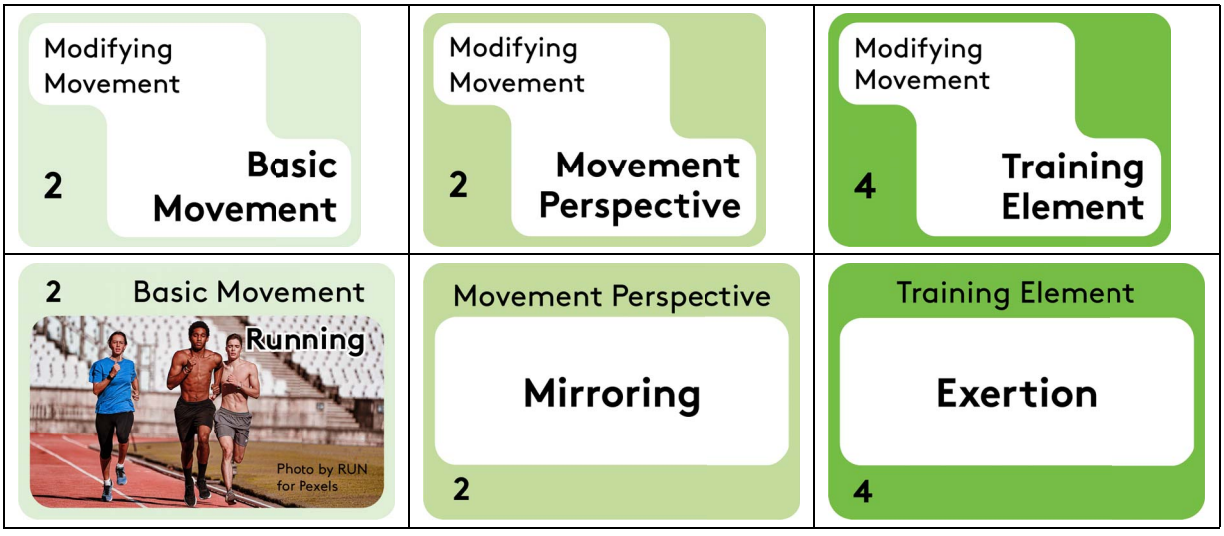


Figure 4. Three different but related Modifier cards: front side of each card on the top and back side on the bottom.

manipulation skills, alongside concepts from dance and coordination (Lubans et al., 2010; Charnavel, 2016), including Rudolf Laban's BESS model (McCaw, 2012), which emphasizes the body in terms of effort, space, and shape, fostering creativity in intentional movement. Modifying Structure draws from sports, games, and play theories to shape movement activities. Informed by game mechanics (Kim, 2015) and play practice concepts (Karoff, 2013), these cards encourage variations of play approaches in competitive or collaborative settings, promoting exploration. The Modifying People category integrates psychological and ecological models, incorporating Jungian archetypes for Persona and environmental affordances to tailor movement based on individual needs. Special needs and constraints are also addressed, ensuring inclusivity in physical activities (Renshaw and Chow, 2019). Modifying Artefact explores technology and game genres to enhance skill development and creativity in movement design, using elements like sensors, points, and leaderboards (Frevel, 2022). Finally, 'Modifying Motive' involves metaphorical and logical elements, stances, values, roles, and perspectives. Drawing on common design practices they incorporate design strategies such as taking on a different persona, adding constraints, and using metaphors (Katan-Schmid, 2017; Elbæk and Hansen, 2019).

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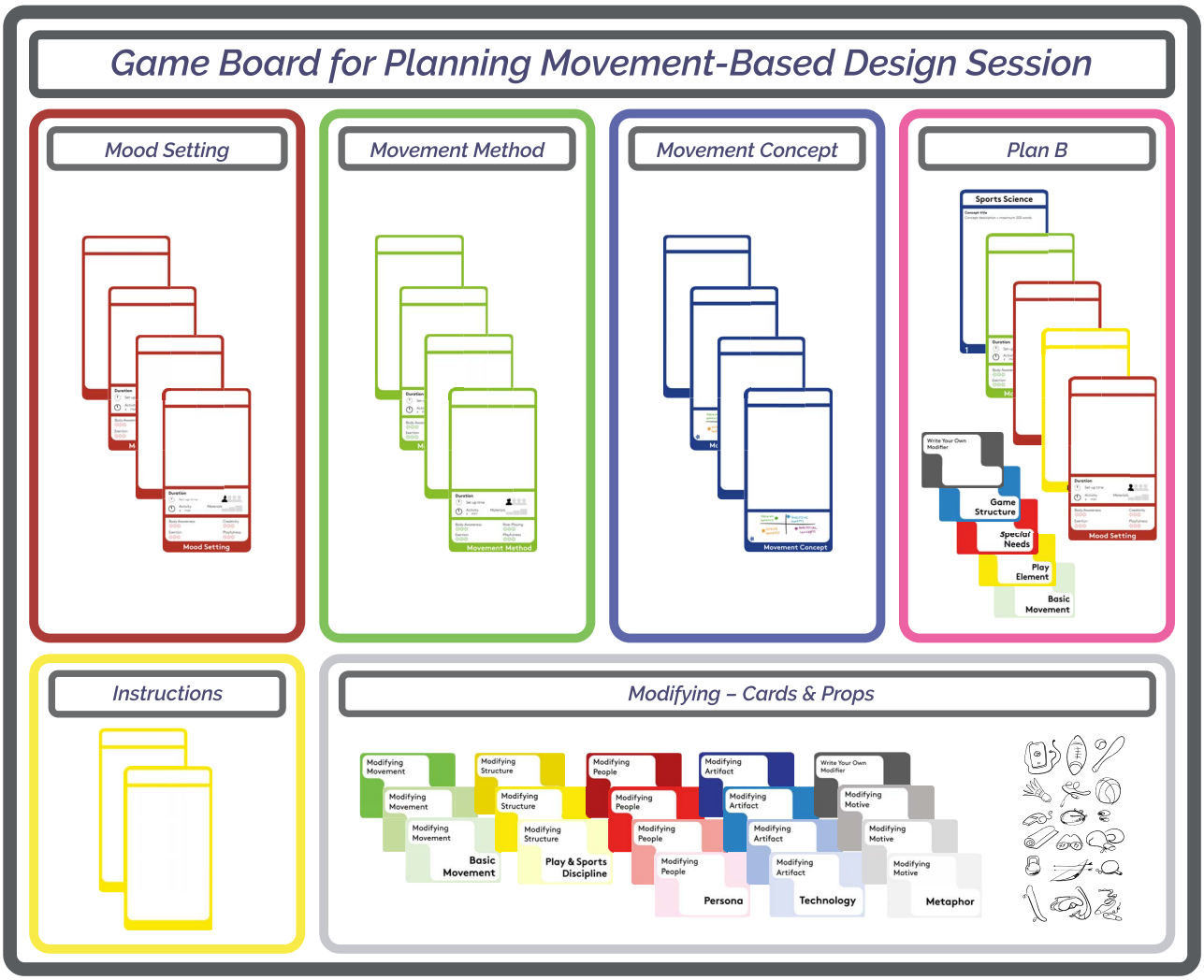


Figure 5. The 'Game Board' with areas to place the various types of cards for a session, including an area for 'backup plan B' cards.

Figure 4 provides three examples of Modifiers, picked from the Modifying Movement category but from three different sub-types (Basic Movement, Movement Perspective, and Training Element). One side of the cards (Figure 4 left) only shows the category and subcategory. The other side of the cards (Figure 4 right) contains a single word, or a word and an image, which suggests a particular change to the current design activity or idea. Each overarching category has its own color, making the cards easier to sort—for example, all of the Modifying Movement cards are in different shades of green. As Modifier cards are intended to be brought into design sessions, they are designed to be easy to use and to contribute as much as possible to keeping participants in a movement flow. The text on modifier cards is restricted to one or two words, or an illustration that is easy to understand. The size of the Modifiers lies between a credit card and a matchbox, making them easy to grasp; their rounded corners make them comfortable to hold and easy to put in a pocket while moving. Since there is a large number of Modifier cards, all cards also have unique numbers, making it possible to sort them and ensure that all cards are back in the box after a session.

In all, modifiers thus provide the potential to creatively explore various movement activities and elements guided by common design perspectives and methods.

3.5 The game board, the design flow cards, and the box

In early trials with the cards, participants and session organizers expressed a need for help in choosing methods and structuring sessions. This need motivated the design of a layout tool (called the 'Game Board') that was designed to support planning and structuring a design session.

The Game Board was inspired by how, while preparing design sessions with early iterations of the method card collection, some facilitators would lay out selected methods, modifiers, and mood setters across a large table to keep an overview of which methods they planned to do next. In particular, one of the researchers acting as facilitator to a design session for music technology had prepared themselves by drawing an elaborate canvas on an A1 sheet of paper. Their canvas had contained dedicated areas delineated for cards that were to be used particular phases, separate areas for stacking modifiers, and an area with backup cards for on-the-fly modification of the session plan. The canvas was furthermore annotated with notes, to remind the facilitator of the goal and intended outcome of each step. This large sheet was then brought into the session to help the facilitator keep track of its structure.

The positive experience of preparing in this way motivated the design of the Game Board (see Figure 5). It consists of a reusable,

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Figure 6. The larger design card box (bottom) containing a smaller box with modifiers, and the modifier box open with the various sets of modifier cards visible (top).

printed, and laminated board with designated areas where concept, mood setter, method, and modifier cards can be placed. It also contains an area for backup and alternative cards. Since the board is laminated, it can be annotated upon using whiteboard markers that can be erased, or alternatively with sticky notes. Compared to the original A1 layout used by the project expert, the board is smaller and relatively simple, supporting ease of access and reuse. For more elaborate sessions consisting of multiple steps, facilitators could conceivably use several separate Game Boards.

The task of planning a design session is furthermore supported by the inclusion of a set of 'Design Flow' cards. These document examples document examples of movement-based design sessions that the project participants had successfully run in previous project, and can thus work both as inspiration, and as templates for constructing design sessions. They suggest how different methods and modifiers can be selected to construct a session, or be varied over different phases in a design project. The Design Flow cards also include brief facilitation guidelines, tips for physical props and DIY material to use, and even suggestions for music. They are not intended to be followed strictly, but rather to provide examples of what others have done.

Finally, the game box itself was designed to provide an easy overview of the cards and the 4 M framework and contains written instructions on how the cards can be selected and used (Figure 6).

3.6 Experiences gathered during the design process

During the design process, the project members carried out numerous design sessions engaging both external participants and facilitators using the methods and the cards. In all, more than 20 sessions were run with more than 300 participants total. Appendix 1 provides an overview of the sessions run as part of the design process.

As can be seen in the overview, the sessions were run for different purposes. Some were run as a way of evaluating methods, or the selection and design of the cards, others were run for the purpose of educating participants in movement-based design methods, and some were run with specific design goals that had been supplied by external participants and professionals. The process included teaching the use of method cards in interaction design courses for master students, and a summer school inviting a wide audience of students and professionals. The cards were also used in design sessions with health professionals, sports and play equipment industry partners, and music education researchers.

Early sessions were done with early versions of subsets of the cards, but gradually more and more of the card set and associated tools were used. Towards the end of this process, complete versions of the toolbox were used, thus exploring the rich variety of ways in which the cards could be selected, sorted, and used before and during sessions. Most sessions were documented through observation notes, video, interviews, and post-session discussions. However, the purposes of the sessions affected what data the participants would consent to provide; and in particular, not all sessions could be filmed.

Towards the end of this process, the participating experts came together to reflect on their experiences of using the cards. To prepare for this reflection they would re-visit their observer notes as well as any previous in-depth analysis of specific sessions (reported in separate publications[3]). During this session, experts would first supply their own experiences as post-it notes. The notes were then sorted and prioritized collectively by the workshop participants. This led to a set of main learnings summarized below.

Most of the card use happens before the actual design sessions . Except for modifiers, the cards had mostly been used when planning sessions. Concept cards in particular have an important role in session planning, in that they help in the selection of methods and structuring the overall session flow (see Figure 7). When brought into sessions (which they sometimes were) they were seldom consulted.

[3] References withheld for anonymity reasons.

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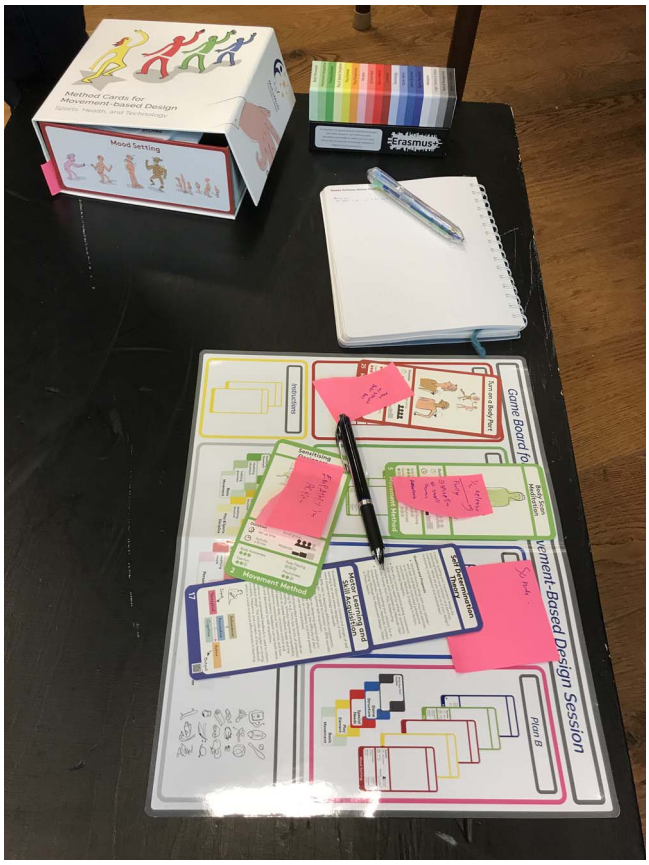


Figure 7. Using the cards and the Game Board while planning a session. Note how some of the method cards have been modified using sticky notes to create bespoke versions of the methods tailored to the design task at hand.

As expected, ***it is difficult to bring the cards into design sessions***. Difficulties relate to their physical handling, as well as to the text on method and concept cards being distracting. These difficulties would sometimes overshadow the benefits of having the cards available: cards were not always brought into sessions at all even though they had been used for planning. As expected, modifiers were much easier to bring into sessions than the other cards.

Game Boards are often brought into sessions, with the cards placed on them. Facilitators do this to keep track of their session plan, as the Game Board allows them to select subsets of cards and arrange them according to their session plan, and very frequently also use the backup area.

Facilitators need flexibility in selecting and adapting methods both during preparation, and on the fly during sessions. The mood setters, which support on-the fly adaptations of methods, and the Game Board with its area for backup cards, help provide flexibility during sessions.

Cards are sometimes handed to participants while running sessions. This is particularly true for modifiers, but sometimes also happens with method and mood setter cards. While this provides opportunities to distribute facilitator tasks among participants, working with the cards also risks break the movement-oriented flow of a session, e.g. if the participants focus too much on reading the card texts.

The card classifications help new users get acquainted with the cards quickly . Classifications such as the level of exertion and if a method fosters body awareness are important for getting acquainted and for providing quick access to alternative methods. However, when selecting cards to structure a session, the text plays a more prominent role as it includes suggestions for methods and modifiers that go well together. Reliance on text contributes to how the cards are primarily consulted during planning.

First-time users benefit from having the cards explained to them by an expert e.g. in a workshop. While the cards stand on their own, practical demonstration reduces the time needed to learn about singular methods as well as understanding how to use the cards.

4 Evaluation with novice facilitators

In addition to the cumulative insights into design sessions throughout the project discussed above, we report here on the experiences of using the card set in a final three-day summer school, organized towards the end of the project, that was focused on teaching movement-based design skills to novice facilitators. The summer school was entirely based on using the card set and toolbox. As such, it presents an excellent context for shedding light on the extent to which the cards provide a basis for learning to use movement-based design methods. It also sheds light on the practicalities of using the cards in designing and running design sessions, especially for inexperienced facilitators.

The summer school had five students who designed and ran their own sessions based on design challenges that the students had brought to the course. Before participating in the course, most had also prepared by joining an online self-study course on Movement-Based Design available from one of the partner universities. Before running their pre-planned session, they had individual tutoring with one of the teachers, who gave feedback on their plan and helped them develop it even further. The sessions ran for 45 minutes and included all stages: warm-up, ideation or sensitizing, and debriefing. Each session had between 5 and 10 participants and included both course teachers and other students. After the sessions had been concluded, the novice facilitators and the participants were interviewed separately about how they had perceived the session.

The design challenges and the students' backgrounds are described in Table 1, together with the cards they used in their design sessions.

4.1 An example design session

S5 was an entrepreneur working in a start-up company where one of the products is an app to promote 'walk and talk' meetings in social life as well as in work practices. At the individual level, this app is intended to counteract loneliness and sedentary behavior, while companies using the app could increase their productivity and innovation and contribute to the well-being of their employees. However, the company noted how users tended to be hesitant both to suggest and to accept participating in walk-talk meetings. This observation was the design challenge that S5 brought to the summer school, and the goal of the session was to explore this hesitation and find ways of overcoming it.

The design session started with a brief introduction to the app and the challenge. Next, a mood setter was run that focused on walking in different ways. The subsequent ideation methods picked for this session were all focused on role-playing different situations with different personas in the work office. Between role-played sessions, the group entered into discussions of how the walk could be structured, such as how the status of the walk as a meeting could be signaled, e.g., by a collective body gesture, and tried out different versions of how a walking meeting could be enacted.

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Table 1. Overview of students, their design challenges, and card use

Student ID	Background	Design challenge	Card use
S1	Background in business studies and human–computer interaction studies at an advanced level	How to help people find their balance in horseback riding	Concept: somaesthetics (not in box) Mood setter: sensitize designers Movement method: collaborative somatic inquiries + modifiers
S2	Bachelor in computer engineering, UX professional designer	How to help drivers adjust to driving on the other side of the street	Concepts: embodied cognition and affordances Mood setter: bubble-gum queen with modifiers Movement methods: 1) Sensitize designers 2) Generate movement from imagery 3) Context bodystorming
S3	Ph.D. student in Interaction Design	Designing for stroke rehabilitation	Concepts: movement learning and skill acquisition, self-determination theory Design methods: body scan Modifier: turn on a body part (mood setter)
S4	Master student in Interaction Design	Game design, ball game	Concepts: none chosen Mood Setters: 1) Tricky B.U.T.T.O.N (variant) 2) 'Detangler' (not in the box) Design method: Forum theatre + empathy in action
S5	Entrepreneur, App developer, dance pedagogue	Encouraging people to do meetings while walking	Concepts: none chosen Mood Setter: Silly Walks with modifiers Design method: forum theatre + modifiers from the Personas set

The facilitator was satisfied with the session and expressed getting out 'a lot' from it based on the original intentions. They were, however, a bit surprised by how much of the session had focused on one specific design suggestion, which had been deconstructed in detail and, to some extent, criticized.

During video analysis, it was noted how the participants spent a fair amount of time and effort in finding ways to signal when the meeting had started. The participants spent time employing the offered methods to better understand and empathize with the problem, so that midway they finally saw the problems that S5 had been aware of when bringing in the case. This highlights how the lack of social practices around walking meetings contributes to the difficulties related to initiating them.

The session participants stated that there might have been too little ideation happening in the session compared to the goals of the facilitator. The participants also noted that one of the design suggestions had been criticized instead of worked through in a constructive way. However, they saw the session as excellent in how it sensitized the participants to issues that the facilitator was already well aware of, and that it would have worked well as a first step towards finding solution strategies.

4.2 Participant feedback

4.2.1 *Feedback on the cards*

There was a consensus amongst participants in the summer school that the cards were understandable and readable and that the facilitators were able to implement the described methods. The sessions were also successful, with facilitators largely able to find suitable Mood Setters, Methods, and Modifiers to fit with their design challenge.

The richness of the card set does lead to stress related to information overflow. For a facilitator with little experience in movement-based design methods, the card set can be experienced as overwhelming. The students at the summer school expressed this sentiment, even though they had had access to the card set online prior to attending the event. We noted how the game board became an essential tool in overcoming this issue. Participants made extensive use of it while planning, picking out sets of cards and trying out different combinations. The final selection, laid out on the board, was almost always brought into the design session and used for easy reference by the facilitator.

Regarding the front side of the cards, the *illustrations* were highlighted as significant reminders of methods. They helped participants sort through the cards quickly and worked as quick reminders of the methods they described, in addition to and separate from the *title* of the method card, which participants thought was harder to remember on their own. Other front-side information, including the number of participants, time requirements and need for tools and props, was less used. However, this information could be more important for experienced facilitators, who already know roughly what they want to accomplish and only need a quick reference to the practicalities of running a particular method or exercise.

One piece of information that students did search for was combinations of cards: Concepts, Mood Setters, Movement Methods, and Modifiers that went well together. While Concept cards include tips on suitable methods and the Mood Setter and Movement Method cards also contain suggestions for suitable Modifiers, this information is included as text and not part of their classification. One student suggested that the cards could be organized into 'suits' of cards that worked well together.

> 'If I have this concept, then maybe I should use one of these methods because they are in the same suit'. (S1).

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It is unclear whether these needs are most articulated during facilitator training while students are learning to use the cards, or if they would also be concerns for more experienced facilitators.

4.2.2 *Reluctance to modify methods*

One thing observed by several students was that they were initially very reluctant to make any modifications to the methods as they were described on the cards. This sometimes led to difficulties for students in finding methods that fit. The teachers were instrumental in helping the students move away from a too-literal reading of the card instructions.

> "In the session with the teacher, I realized that I needed to think about how to use the cards and how to make them fit with my design challenge. Even if I tweaked one method quite a lot, it never occurred to me that I could tweak the other ones!" (S1)

Interestingly, this issue arose even though most of the cards suggest variations of the methods and also include Modifiers that can be used for further variation. In the summer course, teachers instructed in 'letting go' in different ways. For example, one of the teachers suggested that a facilitator could make a new method by combining two cards.

4.2.3 *Learning to instruct*

All of the facilitators commented on how difficult it was to instruct the group on what they had planned to do.

> You have to spend way more time on deciding how you want to explain [the exercise] to the participants (S4)

> I could have given more clear instructions on what they were doing (S2)

Part of the problem lay in the fact that the facilitators did not explain to participants why the session was run and what the intended outcome was. While this was something the facilitators had thought through before the course started, the video analysis shows that most of them did not talk about it at the beginning of their sessions. This might have been intentional:

> Some participants would have needed more explanation on what we were doing, what we were looking for, and why are we doing this. I am kind of split there because part of my thinking was that you were exploring this on your own (S1)

However, being in charge and taking control can sometimes serve as an alternative to giving clear instructions.

> It seemed like the facilitator had a plan, so I just went along with it (Teacher, participating in the session run by S5)

4.2.4 *The importance of setting the mood*

The sessions run at the summer school confirmed the importance of getting the participants into the right mood. For example, the role-playing session (run by S5) was perceived by one of the participants as needing a bit more ice-breaking as they found it awkward to engage in role-playing.

> Going into role-playing was a bit scary if we could have laughed and touched each other (Teacher, participating in the session run by S5)

Most of the sessions selected Mood Setters from the cards available in the box. One session ventured beyond this collection, motivated by how the participants would simultaneously need to play-pretend a competitive game and collaborate in making it more interesting. The facilitator found one mood setter among the cards that was suitable for creating a competitive yet playful mood in the group. To complement it, the teacher suggested a method for creating a close community feeling, which was taken from the 'New Games Movement' (Fluegelman, 1976) collection of playful group games.

4.3 Ideation

As noted already in the introduction, bringing the cards into movement-based design sessions can potentially impact movement and physical action in these. Throughout the project and in the summer school, we observed that cards were primarily used during planning.

To shred further light on this tension point, a reflection, discussion and ideation session was organized to surface the benefits and challenges of bringing the design cards over from planning, into running movement-based design sessions. The session was run at the end of the same summer school in August 2023. The participants of the design sessions were the five students from the summer school and seven experts from the project (some of them also teachers at the summer school). The mix of participants allowed for reflecting both expert and novice perspectives.

The ideation session was divided into three phases. During the first phase, participants were asked to note down (on post-it notes) as many benefits as possible that they found in using the design cards, and then explain them to the others while acting them out physically. The second phase focused on identifying and enacting problems in the same way. The final phase was a tinkering session, in which participants used random materials to construct prototypes for solutions.

All three phases were organized in the same way. First, the participants were given some time to reflect individually. Then, the reflections were presented one at a time. When a particular reflection or idea had been presented, the other participants who had a similar idea or reflection would complement with their perspective. Once a particular theme had been exhausted, another participant would present a new idea. In all three phases, the novices/students provided their thoughts and ideas first, before experts presented their ideas, but all ideas were discussed.

The session was documented by video from two cameras on fixed stands, and additional photos captured by one of the facilitators, documenting the design solutions. The video was analyzed by two of the authors who made notes of all mentions of benefits and drawbacks, and created short descriptions of each of the ideas generated based on both their physical demonstration and the way they were verbally described.

Below, we focus on the design qualities that affected how the participants suggested to address bringing cards into sessions. The participants also provided comments on the use of cards during planning. These related primarily to the ***physicality*** of the cards, their ***organization***, and their ***graphical*** and ***physical*** design; the comments overlap to some extent with the comments discussed in the evaluation section above. They are omitted in this section unless they have relevance for bringing cards into sessions.

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5 Results from ideation

5.1 Benefits

As expected, the method and mood setter cards were considered more useful for planning sessions than in the actual sessions. However, their role as ***reminders*** stood out as important for sessions. All participants had used these cards extensively during planning, sifting through them and organizing them in different ways, and sometimes adding sticky notes to tailor methods further. These organizations were maintained as reminders during sessions, but primarily left untouched, e.g. kept at a table at the side of the area used for the movement exercises. The ***Game Board*** was essential for this function, as the layout of cards on the game board worked as a visual reminder of the current activity, and the overarching structure of the planned session. The participants mentioned how they had found it useful to make notes and attach sticky notes to the board as well as to cards.

In their role as reminder, the ***images*** on cards were mentioned as key. One of the students reported that when sorting through the methods and modifiers, they had selected the image first and then read the descriptions. Participants also learned to recognize them during planning, and needed not re-read their text in the actual sessions.

The card texts became useful when methods contained detailed instructions to the participants. For example, one facilitator had read these aloud from the card.

In line with previous work (Elbæk et al., 2023; Rasmussen et al., 2023; van Rheden et al., 2024), the ***modifier*** cards were seen as providing flexibility as they could be selected on the fly to adapt to the ongoing activity. Participants also had also used them to create chance interventions through shuffling them in advance. One participant commented that a good chance effect could be to throw the modifiers into the air, to then pick out some of them.

It was also noted that ***the physicality of the cards was important in organizing and running sessions***. Facilitators used cards as signals: holding a card indicated leadership and created focus. Facilitators would also pick up new cards to signal that something new would happen.

Participants also appreciated how the modifier cards could be ***distributed among participants*** as a way of distributing tasks, enabling more participatory action. A particularly interesting example was how one of the students, in their design session, had invented a movement-based design game based on the mood setter cards. In the game, the mood setters were put in a pile at one end of the room. The participants were tasked with throwing balls into a bucket. When somebody managed to hit the bucket, they were allowed to pick up a modifier card, and then reinvent the game based on the modifier. This is an example of how the cards can be integrated in the physical and social setup of a design session, contributing to movement engagement as well as ideation.

5.2 Drawbacks

The participants considered the main issue with using the cards in design sessions was that they are ***hard to handle while moving***. Since the Game Board was perceived as useful in sessions, the fact that the cards could not be stuck to the board also became an issue. One of the teachers commented on their experience of using the cards outdoors, and how the cards had blown away in the wind.

A slightly less obvious drawback was that ***the cards draw visual attention***. As discussed previously, this was sometimes used by facilitators in e.g. holding a card up to signal a change. But more commonly this was considered a problem, in that it steered the gaze of participants and facilitators hampering free movement. The fact that facilitators sometimes would stop and read the cards was considered particularly disruptive for maintaining the flow and mood of sessions.



Figure 8. A version of the Game Board enhanced with 'pockets', allowing cards to be secured to it.

The sheer ***number of cards*** would also sometimes become a problem. This problem mainly surfaced during planning. Some facilitators had the experience of bringing a large number of modifiers into sessions and then having trouble finding the right one.

Finally, some comments related to that much of the information that participants used to structure their sessions was ***textual***, such as when method cards contained suggestions for mood setters and modifiers that would work well together with the method. Apart from the modifier texts, the text on the cards was considered too small to be useful in sessions. The participants suggested that some of the information could instead be color-coded. This would help in sorting and searching for cards during planning, and also work to provide quick overviews when running sessions e.g. over different phases of a design session.

6 Tinkering suggestions for handling cards while moving

The tinkering session was restricted to what was perceived as the main issue with bringing cards into sessions: the handling of cards while moving. A large box of design materials was brought in to support the participants in creating rough prototypes for solution ideas. The participants also used both the cards and the Game Board as props. When they had finished constructing some solutions each, the participants enacted their ideas to the whole group, and some more variants were suggested by the group. In all, 16 ideas were generated this way, of which 8 were generated by the students and 8 by the experts.

Due to the focus on how to handle the cards, there was limited variation among the ideas. Post-session analysis identified four main solution strategies, which sometimes were combined in one and the same design idea.

Attaching cards to the game board. This was the most common idea. Both students and teachers produced different variants of a 'sticky' game board that would be easier to bring into the sessions (see e.g. Figure 8). It was mentioned that the function is easily achieved through pasting the cards to the Game Board using adhesive pulp. One person suggested that both sides of the Game Board should be printed, to provide a higher level of flexibility in running sessions.

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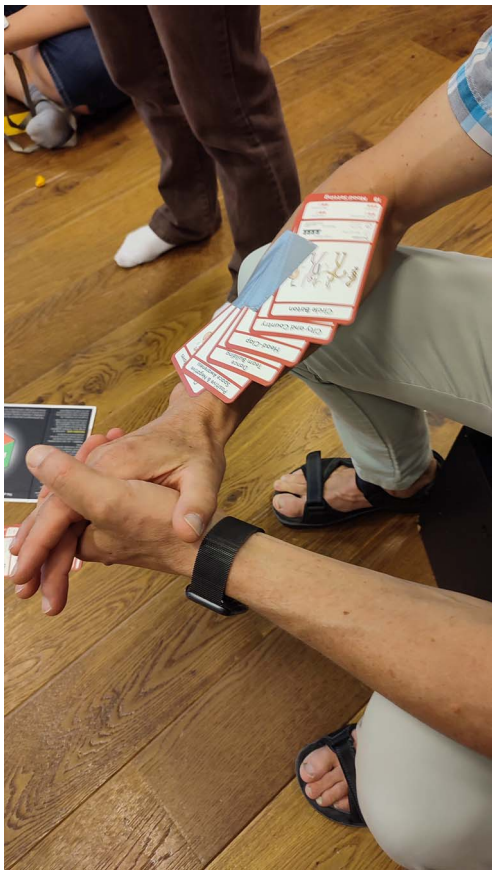


Figure 9. A wearable timeline attached to the facilitator's wrist, from which the cards can be discarded one at a time.

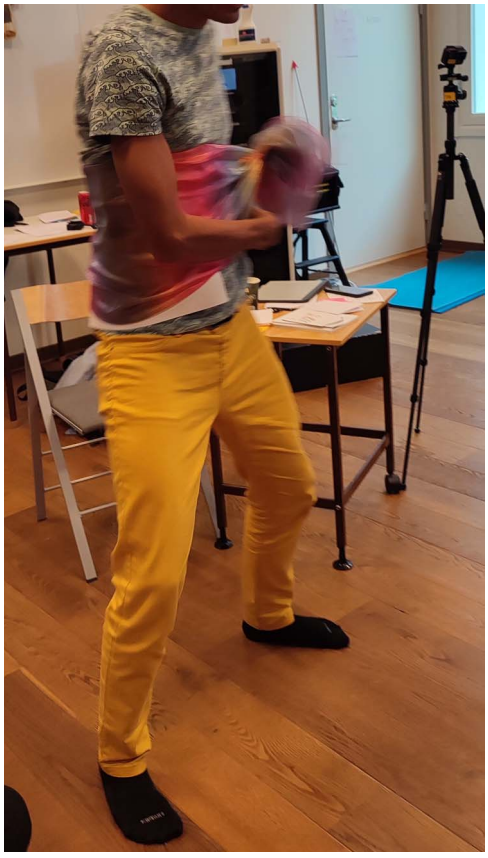


Figure 10. Attaching the whole Game Board to the facilitator's body by folding it into a piece of cloth and wrapping it around the torso.

These solutions maintain the role of the Game Board as an overview and structuring tool. They also allow the facilitator to move freely as most solutions did not involve carrying the cards that were instead kept at the Game Board. The use of the game board in sessions creates a focal point in the space, located where the Game Board (and therefore, the session plan) resides. This can be both hindering and helping.

Wearing the cards on the body. This strategy focused on making it easier for the facilitator to keep track of the cards. Solutions included pockets that could be strapped onto the body, wearing the cards on the arm (Figure 9), and punching a hole through the cards to be able to carry them in a key ring. It was perceived as beneficial when the facilitator could keep the cards close in ways that did not interfere with actions and movement. These ideas tended to make the cards private to the facilitator, making it less likely that the facilitator would use them for sharing tasks with the participants.

While these solutions solve the problem of letting the facilitator move, they did not provide the same level of overview of sessions that the Game Board afforded. A notable exception was a solution suggested by one of the students for strapping *the whole Game Board* to the body of the facilitator (Figure 10) (making it visible for the participants but less legible for the facilitator).

Creating a timeline from which the cards could be discarded when used. These solutions were combined with some way of fastening the cards, either to the facilitator's body, or some tool (such as a stand or a string, see Figure 11). The prevalence of these suggestions indicates that the complete overview and flexibility that the Game Board provides might not be necessary in session, but that a simpler timeline would provide enough of an overview. The participants that had created these solutions also considered it important to be able to physically discard the cards, as this could create a shared understanding among the group of how the session was progressing. Hence, a simpler timeline visualization might contribute to making session planning more participatory than by using the full Game Board.

Changing the format of cards used in sessions. While most ideas used the cards as-is, a few solutions suggested that the cards themselves could be redesigned to be easier to bring into sessions. One student and one expert suggested that the cards could be digital, e.g. made available on a wristwatch worn by the facilitator.

One of the teachers suggested that the images on method and mood setter cards could be printed separately, as small cards would be more suitable for bringing into sessions and it was mainly the images that were used in sessions. This would require that the images would be detachable from the main cards. The full cards would be used during planning, while the detached picture would be the only thing brought into an ideation session.

Finally, several ideas focused re-creating the mood setters as objects instead of cards. This involved attaching the mood setters to something else, e.g. by sticking them onto objects to create physical representations of their concepts, or simply to make them easier to throw around the room (Figure 12). These ideas support sharing cards among participants and hence contribute to participatory approaches to session management.

7 Discussion

In comparison to many other ideation cardsets, a particularity for this cardset is that the cards document, and help navigate and

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Figure 11. Two variants of timelines: cards attached and ordered on a rope (left) and cards attached to sticks that can be placed in a holder or in the ground in the right order (right).

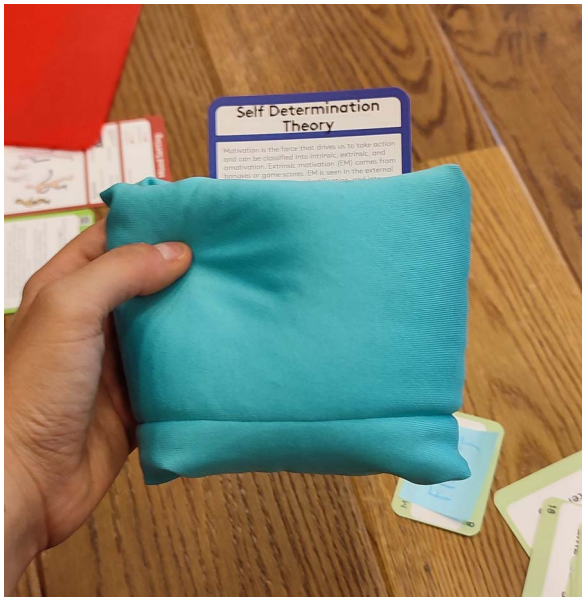


Figure 12. A soft pocket for cards making it possible to throw them across the room.

select, methods, rather than just support their execution. It is primarily the mood setters that fulfil functions similar to those of ordinary design method cards, in that they provide challenges and inspiration for the ideation process itself. The other cards mostly help planning and structuring ideation, by providing easy access to methods and finding suitable combinations of methods (Rasmussen et al., 2023).

We indeed found mood setter cards to be useful resources in sessions. Their physical form, as small but sturdy cards with limited content, was furthermore instrumental to their function in that it allowed for multiple ways to use them in-session.

Moreover, our experiences show that the design method and mood setter cards, cards that document the methods and exercises and for that reason have their main use in planning, also have their use in sessions. In this role, they primarily worked as reminders for the facilitators. This can be contrasted with how previous literature on movement-based design typically assumes that facilitators know their methods by heart, and that this skill is important to train, see e.g. (Höök et al., 2016a, 2016b). In line with (Rasmussen et al., 2023), we found that facilitators can learn methods from cards. We believe that facilitators benefit from having access to a wide collection of methods and exercises during planning. Selecting among a wide range of methods allows them to tailor their session to the precise design challenge at hand, and gives them flexibility in combining or adapting them using e.g. mood setters or sticky notes. But this also means that facilitators may not know all the methods by heart, and hence, bringing the method and mood setter cards into sessions will sometimes be necessary.

Having access to a wide range of methods and ways to tailor them to a precise design problem also means that each design session can be unique. Hence, the method and mood setter cards are needed in sessions also to keep track of the structure of the session. Such reminders must be given a format that fits with the sociospatial structure. After all, movement-based design is dependent on the space, the design tools and materials at hand, as well as the movement and orientation of the participants. From using the cards in a variety of contexts we saw that the cards have multiple properties that make them suitable for this purpose, such as their color-coding, images, and that they can be physically arranged to provide meaningful overviews e.g. using the Game Board as a basis. The design suggestions made by our workshop participants further emphasized the importance of keeping cards close to the body in ways that resemble how people carry tools in a toolbelt, as well as supporting physical arrangements of the cards e.g. to maintain their time line.

An open question is if there are alternative formats that would have worked even better. The cards are text-heavy, especially the movement concept cards, and text was (with few exceptions) not useful in ideation sessions, and could hamper the flow of

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sessions when insecure facilitators spent time reading it. From this perspective, the design suggestion to make the images of cards detachable is particularly interesting as it would create two versions of the cards: one full card used during planning, and a small version of the card only containing the image as a reminder.

It could also be possible to use interactive design objects instead of cards. An example of this is how (Biggelaar, 2023) investigated the possibility of replacing or complementing modifier cards with digital-physical objects that facilitators would bring to design sessions. These interactive objects inspired designers to move in new ways with the objects, such as wiggling, rocking, spinning, and fidgeting with the objects.

8 Conclusion

Movement-based design is gaining popularity in the field of Interaction Design, especially in the context of sports, movement, and well-being. In this paper, we share practical insights from using method cards to enrich movement-based experiences in design sessions, based on a publicly available collection of method cards[4]. We found that our method cards are mostly used in preparation of movement-based design sessions with facilitators placing the method cards on the accompanying Game Board to plan the flow of the design session. However, we found that while cards can be difficult to bring into sessions due for practical reasons, they also offer benefits when brought into sessions such as offering opportunities for distributing facilitation-related tasks among participants.

An ideation session was conducted to elicit strategies to overcome the practical difficulties related to card management. The ideation session uncovered multiple strategies for managing the cards that can be supported by simple tools.

We conclude that our card set is of practical use in sessions, and supports the precise tailoring of design sessions to the design task at hand.

Conflicts of interest

None declared.

Data availability

The data underlying this article is of qualitative nature and cannot be shared publicly as it cannot be securely anonymized. The data will be shared on reasonable request to the corresponding author.

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Appendix 1

Overview of design sessions. The list is not chronological as sessions were run by multiple partners for varying purposes.

#Participants	Who participated	Who facilitated	Were the cards used?	Were the cards brought into the design session	What was the design goal/theme of the session	What data were collected?
~**30**	Game design students and researchers	Two teachers. One external, one from project	Only methods	n/a	Open-ended sensitizing activities for game designers and researchers.	Video recording of session and interviews. Post-ideation interviews with participants
29	Students in Interaction Design	Three external teachers	Mood setters	Cards were left by each game station for facilitators and participants to consult at their own discretion.	Evaluation of the cards	Questionnaire for facilitators Observer notes.
8	Master students in HCI	Two external teachers	Full set of cards	Modifiers were used during the session	Open-ended trial of the modifier cards	Post-questionnaire for facilitators, observer notes, and open interview with facilitators and participants Video recording
22	Bachelor and master students with backgrounds in game development, education, and sports sciences	Two project members	No	n/a	Early trial of game-based methods to be included in the design methods set	Pre- and postquestionnaires, open interview, and observer notes
80	University students in K-12 PE education	10 facilitators: 4 students 6 PE high-school teachers Support facilitation from project	Yes	Method cards read once or twice. Preselected modifier made available during the session	Three tasks focused on different target groups for design: kindergarten kids; eight graders 20 min PA in school; seniors in activity center	observation notes, interviews with seven of the facilitators
65	PE sports university students in PE sports education	Teacher from project	Yes	Modifiers	Training sports science students in design and facilitation skills	Oral feedback from students in debrief
34	Sports industry participants	Two facilitators from project	Yes	Modifiers	Testing the MeCaMInD toolbox with sports industry partners	No
23	Sports industry participants and students from a para-sport school	Three facilitators from project	Yes	Modifiers	Ideation of inclusive exergame for impaired and non-impaired people to play together	Documented presentation of the ideation result
35	Sports industry participants with a specific focus on workplace physical activity	One facilitator from project	Yes	-	Ideation of functionalities in a 50-minute app supporting workplace physical activity	Documented presentation of the ideation result
39	Sports industry participants with a specific focus on youth football players mental health	One facilitator from project	Yes	-	Ideation and goal clarification regarding the potential for designing an app supporting mental health among youth football player	Documented presentation of the ideation result

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#Participants	Who participated	Who facilitated	Were the cards used?	Were the cards brought into the design session	What was the design goal/theme of the session	What data were collected?
16	Innovation and Entrepreneurship Sports Science students	Two facilitators from project	Mood Setters and Modifiers	Modifiers	Training of sports science students design and facilitation skills, Design outdoor event for school students 10 to 14 year	Students wrote white papers on the event design process
6	Professionals in physiotherapy and dance	Project participants	Yes	Unclear	Investigating the applicability of the MeCaMInD cards in instrumental and normative domains. Groups worked with scenatios from their professional practices	Video recordings Observer notes Pre-and post-session interviews
3	Performance team: three choreographers, design researcher; art tech expert; music composer	The three choreographers +2 project participants	Only methods	n/a	Several sessions, run with the goal of creating a live performance with dances representing an immersive and multisensory network.	Debrief conversations, recorded
20–25	Students MSc design/technology	Two project participants	Yes	Modifiers (not yet in card format)	playground setting + field next to it; ballroom + square next to it	Students' postparticipation reflections on the course and project
6	Interaction designers and urban landscape architects	One project participant	Yes	Method cards as reminders, Modifiers	Playground design	Video recordings, interviews
30?	Sports professionals and teachers	Two project participants	Yes	Modifiers	design activities based on a particular piece of technology	Brief notes by facilitator
7	Music therapists and music therapy teachers/researchers	One project participant	Yes	Both method cards and modifiers brought into session	design technology to improve impro-music therapy sessions for emotion management	Extensive notes by facilitator
29	Students learning movement-based technology	Two project participants	No	No	Focus on designing VR experiences	Video-recording of physical part of the session Screen capture of the virtual part Focus group interviews Student reflections
15	MSc students HCI	Two project participants	No	No	VR bodystorming	No documentation, preparatory session
5	Visitors at a project dissemination workshop	One project participant	Only methods	No	Design for restraint/constraint; special user group	Some personal reflection notes
	IxD master-level students	Three project members	Yes	Modifier, physical prototype	The theme of the ws was 'dealing with procrastination'	Video recording Survey Fieldo notes with photographs
8	Physiotherapists and researchers in interaction design	Two project members	Only methods	Modifiers	Technology to help guide and correct physiotherapy exercises	Collection of design ideas Recorded discussion of the ideas

(Continued)

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#Participants	Who participated	Who facilitated	Were the cards used?	Were the cards brought into the design session	What was the design goal/theme of the session	What data were collected?
5	Physiotherapist, students	Two project members	Only methods	No	Selection and adaptation of methods for a later and larger session	
8	Physicotherapists and researchers in interaction design	One project member	An early version of the cards used in planning the session	No	Investigate the use of interactive artefacts in the physiotherapy practice.	Documentation of the outcomes of the design session
11	Students at the MeCaMind summer school + MeCaMinD participants	Three project members	Yes	Yes, as object of investigation	Evaluate the cards with respect to their physicality, and bodystorm ideas for alleviating found issues.	A discussion on what methods had been used, and why. Video recording
16	Students at the summer school + project participants	6	Yes	All types	Train the summer school students in facilitation and in using the cards. The students also brought their own design challenges.	Video recording of design sessions and debriefs. The student facilitators did a debrief as part of their session. The facilitators AND participants were interviewed separately about their experiences of the session.

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